Jaya Sai Krishna Muppalla

Staff Software Engineer

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SUMMARY

Staff Software Engineer with 11 years designing and operating distributed systems and cloud infrastructure for Samsung Electronics at consumer scale. Builds Kubernetes platforms, cloud-native backend services, streaming data pipelines, and applied AI/RAG systems, and owns the CI/CD, observability, and on-call practices behind them.

SKILLS

Languages & Frameworks: Java, Node.js, TypeScript, Go, Python, Spring Boot, Express, NestJS, C/C++, SQL

Cloud & Infrastructure: AWS, Azure, OCI, Kubernetes, Helm, Docker, KubeVirt, Terraform, Serverless Framework, Packer

Data & Messaging: Kafka, Kinesis, Apache Flink, MongoDB, Redis, MySQL, Elasticsearch

Observability & CI/CD: OpenTelemetry, CloudWatch, PagerDuty, Prometheus, Grafana, Traefik, GitLab CI/CD, GitHub Actions, Jenkins, k6

APIs & Protocols: REST, GraphQL, WebSocket, gRPC, MQTT, WebRTC, RTSP, SCTE-35, VAST

AI & Data Retrieval: Agentic AI, Claude, Amazon Bedrock, Codex, LangChain, Azure OpenAI, OpenAI API, Retrieval-Augmented Generation (RAG), Vector Search

PROFESSIONAL EXPERIENCE

Samsung Electronics America

July 2019 – Present

Staff Engineer, Back-End Server

Feb 2025 – Present

- Built an in-house linear channel origin for Samsung Smart TVs' FAST (free ad-supported streaming TV) service on Node.js/NestJS and Kubernetes (EKS), in place of AWS Elemental MediaTailor Channel Assembly, assembling looping and scheduled playlists for ambient channels into continuous HLS streams through a CDN-backed origin.
- Implemented server-side ad insertion (SSAI) using SCTE-35 markers, and a VAST-compliant ad service that routes Samsung TV ad inventory to interactive mini-games served as playable ads.
- Designed a real-time ad-event analytics pipeline (AWS Kinesis, S3, Glue, Athena) that captures VAST tracking events, with QuickSight dashboards for playable-ad performance reporting.
- Built a custom Kubernetes controller to reconcile published channels in MySQL with Kubernetes custom resources (CRDs), which automates channel infrastructure management.
- Developed a GPT-based movie recommendation system using Retrieval-Augmented Generation (RAG) with LangChain and Azure OpenAI, and vector search using Azure AI Search and Redis.
- Set up Claude and Codex on Amazon Bedrock for internal developers, and onboarded engineering teams onto them.
- Owned a seven-version, zero-downtime Kubernetes upgrade (1.26 to 1.33) with a staged rollout, per-step validation, and a documented rollback runbook.
- Migrated observability to OpenTelemetry for a single instrumentation standard across logs, metrics, and traces, with CloudWatch dashboards and PagerDuty on-call routing built on top.
- Wrote k6 performance tests and automated their runs, with custom reports and alerting.
- Reduced AWS infrastructure costs through instance rightsizing and workload-aware instance selection, automated cleanup of idle resources, and team-level cost visibility with budget alerts.
- Reduced internal network egress costs by 75% by migrating service traffic to AWS VPC endpoints and updating route tables.
- Own CI/CD and production operations for these systems. Every release reaches production through Terraform and Helm, and the remediations that page most often are automated away.

Engineer 3, Software Process Engineer March 2022 – Jan 2025

- Automated GPU-backed virtual game machine provisioning with KubeVirt (Kubernetes + VMs) across AWS and OCI bare-metal, running games on NVIDIA and AMD GPUs via vGPU/SR-IOV on Windows and Linux.
- Automated optimized EKS/OKE machine images for game streaming using Packer (qcow2 format), including host GPU driver management.
- Added a custom Kubernetes host-port operator to make game machines externally reachable at lower cost, and containerized Windows games to run on Linux using Wine.
- Stood up and ran a Rook + Ceph storage cluster for Read-Only-Many (ROX) persistent volumes, which AWS did not support natively.
- Extended the Puppetlabs PVPool controller with Kubernetes volume snapshots, which cut PVC replication for game data to near-instant.
- Delivered sub-100ms end-to-end latency for interactive cloud gameplay.
- Led a 3-man DevOps team for the GameHub O&O platform, directing two engineers in architecting cloud infrastructure for in-house gaming strategies

Engineer 2, Software Process Engineer July 2019 – Feb 2022

- Built and operated Node.js backend services on Kubernetes (AWS EKS), deployed with Helm, supporting Samsung wearable B2B applications for health monitoring, beacon-based indoor positioning, proximity detection, and contact tracing.
- Worked with wearable-generated health and biometric data, including heart rate (HR), heart rate variability (HRV), accelerometer signals, sleep metrics, and location data, supporting the ingestion and processing of sensor data from wearable devices.
- Used Apache Flink and Complex Event Processing (CEP) to process real-time event streams and detect patterns across wearable sensor and location data.
- Contributed to security practices for infrastructure and sensitive health and biometric data, aligning system design and operational controls with HIPAA requirements, SOC 2 controls, and NIST security frameworks.
- Designed event-driven, serverless microservices on AWS using a Function-as-a-Service (FaaS) architecture, with deployment automation through the Serverless Framework.
- Developed reactive backend services and GraphQL APIs using Java, Spring WebFlux, MongoDB, and Kafka to support scalable data access and event-driven application workflows.
- Engineered multi-region AWS infrastructure with Terraform to support availability, security, and regulatory requirements.

First Object Inc **Aug 2015 – July 2019**

Server Engineer

- Client: Samsung Electronics America.
- Developed and deployed serverless applications across multiple global AWS regions to minimize latency, with geospatial queries and geohashing powering geofence and location-based alerting.
- Integrated mapping APIs including OpenStreetMap, Leaflet, Google Maps, and HERE Maps for geocoding and location-data visualization.
- Built Spring Boot microservices and IoT applications using the MQTT protocol for device communication.
- Designed a reactive, real-time UI using Meteor, Handlebars, and MongoDB, and contributed across the stack including CMS (Magnolia) and Kafka streaming.

EDUCATION

Master of Science, Computer Science, University of Missouri Kansas City (UMKC) **Aug 2013 – May 2015**

Bachelor of Technology, Electronics and Communication Engineering, Jawaharlal Nehru Technological University **May 2009 – April 2013**